Т4. $\xi \sim \rho(x)$

$$\int_{-\infty}^{+\infty} dF(x) = 1$$

$$\rho(x,\theta) = \theta\{(-1,0) \vee (0,1)\} + (0,5-\theta)\{0\} +$$

$$+ (0,5-\theta)\{2\}$$

① $$M\xi = \int_{-1}^{1} x\,dF(x) = \int_{-1}^{1} x\theta\,dx + 2(0,5-\theta) =$$

$$= 1 - 2\theta$$

$\tilde{\theta} = 0{,}5(1-\bar{X})$

$M\tilde{\theta} = 0{,}5M[1-\bar{X}] = 0{,}5 - 0{,}5(1-2\theta) = \theta$

$\tilde{\theta}$ - несмещ.

$M\xi^2 = \int_{-1}^{1} x^2 \theta dx = 2 - 4\theta = 2 - \frac{10}{3}\theta$

$D\xi = 2 - \frac{10}{3}\theta - (1-2\theta)^2 = 1 + \frac{2}{3}\theta - 4\theta^2$

$D\tilde{\theta} = \frac{1}{4}D[1-\bar{X}] = \frac{1}{4n} + \frac{\theta}{6n} - \frac{\theta^2}{n} \xrightarrow{n\to\infty} 0$

$\Rightarrow \tilde{\theta}$ состоят.

m - кол-во раз в выборке попалось 0 или 2

тогда

$L(\theta) = \theta^{n-m}(0{,}5-\theta)^m$

$\ln L(\theta) = \ln\theta^{n-m} + \ln(0{,}5-\theta)^m =$

$= (n-m)\ln\theta + m\ln(0{,}5-\theta)$

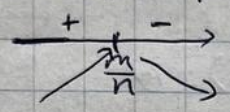
$\frac{\partial \ln L}{\partial\theta} = \frac{n-m}{\theta} + \frac{m}{0{,}5-\theta}\cdot(-1) = 0$

$\frac{n-m}{\theta} - \frac{m}{0{,}5-\theta} = 0$

$(n-m)(0{,}5-\theta) - m\theta = 0$

$2n\theta = n - m$

$\tilde{\theta} = 0{,}5 - 0{,}5\frac{m}{n}$



$\Rightarrow$ m. max

$p = P(\xi = 0) + P(\xi = 2) = 1 - 2\theta$

$M[\frac{m}{n}] = 1 - 2\theta$

$M\tilde{\theta} = 0{,}5 - 0{,}5(1 - 2\theta) = \theta \Rightarrow \tilde{\theta}$ – несмещ.

$D\tilde{\theta} = \frac{1}{4} D\left(\frac{p(1-p)}{n}\right) = \frac{1}{n} \cdot \theta(0{,}5 - \theta) \xrightarrow{n \to \infty} 0 \Rightarrow$

$\tilde{\theta}$ – сост.

• Проверка эффективности

$- \int_{-\infty}^{+\infty} dx \frac{\partial F}{\partial \theta} = \int_{-1}^{1} dx \; = 0$

$\frac{\partial}{\partial \theta} \int_{-\infty}^{\infty} dx F(x, \theta) = 0$

$- I(\theta) = \int_{-\infty}^{\infty} \left(\frac{\partial \ln(p(x, \theta))}{\partial \theta}\right)^2 dx F(x, \theta) =$

$= \int_{-1}^{1} \frac{1}{\theta^2} \theta \, dx + \frac{1}{(0{,}5 - \theta)^2} (0{,}5 - \theta) + \frac{1}{(0{,}5 - \theta)^2} (0{,}5 - \theta) =$

$= \frac{2}{\theta(1 - 2\theta)}$

– $p(x,\theta)$ непр дифф при $\theta \in R$

$I(\theta) > 0$

$I(\theta)$ - непр на $(0, \frac{1}{2})$ => модель регул.

$\tilde{\theta}$ несм, ВМ регул, $D\tilde{\theta} = \frac{\theta(0.5-\theta)}{n}$ - огр на $\forall$ комп. на $(0, \frac{1}{2})$ => $\tilde{\theta}$ - эфф.

• ОМП

$\tilde{\theta}$ несм, ВМ регул, $D\tilde{\theta} = \frac{\theta}{6n} - \frac{\theta^2}{n} + \frac{1}{4n}$ - огр на $\forall$ комп. из $(0, \frac{1}{2})$

$$D\tilde{\theta} > \frac{\theta(1-2\theta)}{2n}$$

$$\frac{1}{4n} + \frac{\theta}{6n} - \frac{\theta^2}{n} > \frac{\theta(1-2\theta)}{2n} = \frac{\theta}{2n} - \frac{\theta^2}{n}$$

$$\frac{1}{4n} + \frac{\theta}{6n} > \frac{\theta}{2n} - \cancel{\frac{\theta^2}{n}} + \cancel{\frac{\theta^2}{n}}$$

$$\frac{1}{4n} > \frac{2\theta}{6n}$$

$$\frac{1}{4n} > \frac{\theta}{3n}$$

$$\theta < \frac{3}{4}$$

=> оценка неэфф.